26-2-22 Astronomy and Cosmology Saturday

Luminosity (L)

Energy emitted by the star in unit time

OR

Power Emitted by the star

$$\text{Luminosity} = \frac{\text{Energy}}{\text{time}}$$

Unit => Watt = Js^{-1}

* Luminosity of star depends on 2 things
  a) Temperature of the star
  b) S.A of the star / Size of star

→ (Temperature of star)^4 ∝ Luminosity
→ Size / S.A of star ∝ Luminosity
} Directly propotional

Luminosity ∝ (temp)^4

$\Rightarrow L \propto T^4$

Luminosity ∝ Area

$L \propto A$

if we combine both the eqn

$L \propto AT^4$

$L = (\text{constant})\, AT^4$

↓

stefan boltzmann constant

$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$

↓

$$\boxed{L = \sigma AT^4}$$

* Luminosity is a property of star

* we assume that all stars are perfect sphere so we can apply the formula of area of sphere to get area of star

* Now we will [illegible] luminosity but for sun we associate "Solar luminosity"

Solar Luminosity (L_o)

Luminosity of sun is called "Solar luminosity"

OR

Energy emitted by the sun in unit time

OR

Power emitted by the sun

$L_o = 3.83 \times 10^{26}\ W$

→ An object of known luminosity is called "Standard Candle"
e.g sun is also a standard candle

Standard Candle

An object of known luminosity is called "Standard Candle"

e.g any star of known luminosity can be said as standard candle

Types of Stars

1- Cepheid Variable star
2- Type 1A supernovae

* Energy emitted per unit time by star is called "luminosity of star" At surface of star

* Energy received per unit time on earth is called "Radiant flux density" or "brightness"

Cepheid Variable Star

A star whose brightness varies periodically is called "Cepheid Variable star"

* A star whose brightness ~~is~~ varies with equal interval of time

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Type 1A Supernova

A star which implodes rapidly with release of matter & energy is called "Type 1A supernova"

→ Brightness is constant

* We measure the distance of star from earth in terms of light years
* light year is a unit of length / distance

Light Year (Ly)

Distance travelled by light in vacuum in one year

Ly = Distance travelled by light in one year

= speed × time

* Star is producing this light

one year

$= (3 \times 10^8) \times (365 \times 24 \times 3600)$

$$\boxed{Ly = 9.5 \times 10^{15}\ m}$$

Example

Distance of Sun from Earth => 1.5×10^{11} m

" " " " "

in light year => $\frac{1.5 \times 10^{11}}{9.5 \times 10^{15}}$

=> $\boxed{1.8 \times 10^{-5}\ ly}$

Distance of Proxima Centauri => 4.2×10^{16} m

in light year => $\frac{4.2 \times 10^{16}}{9.5 \times 10^{15}}$

= $\boxed{4.2\ ly}$

The [illegible] at the surface of Earth depends on 2 things ;-

a) Brightness ∝ luminosity

b) Brightness ∝ $\left(\frac{1}{\text{distance}}\right)^2$

Combine the two eq,

$$\text{Brightness} \propto \frac{\text{Luminosity}}{(\text{distance})^2}$$

$$\text{Brightness} \propto \frac{L}{d^2}$$

$$\text{Brightness} = (\text{Constant})\ \frac{L}{d^2}$$

↓

$\frac{1}{4\pi}$

$$\text{Brightness} = \frac{L}{4\pi d^2}$$

also called

$$\boxed{\text{Radiant flux Intensity} = \frac{L}{4\pi d^2}}$$

Radiant flux Intensity

Energy of star passing perpendicularly per unit area per unit time is called "Radiant flux Intensity"

OR

Power of star passing perpendicularly per unit area

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Radiant flux Intensity $= \dfrac{\text{Energy}}{\text{Area} \times \text{time}}$

OR

Radiant flux Intensity $= \dfrac{\text{Power}}{\text{Area}}$

$$F = \frac{\text{Power of Star}}{\text{Area of Star}}$$

$$\boxed{F = \frac{L}{4\pi d^2}}$$

Unit = Wm^{-2}

Q.1

Luminosity of Sun => 3.83×10^{26} W

Radius of Sun => 6.96×10^{8} m

Distance of Sun from Earth => 1.5×10^{11} m.

Radiant flux intensity => ?

i- At surface of Sun

ii- At surface of Earth

$$F = \frac{L}{4\pi d^2}$$

Ans i) $F = \dfrac{3.83 \times 10^{26}}{4 \times 3.142 \times (6.96 \times 10^{8})^2}$

$$\boxed{F = 6.3 \times 10^{7}\ Wm^{-2}}$$

ii) $F = \dfrac{3.83 \times 10^{26}}{4 \times 3.142 \times (1.5 \times 10^{11})^2}$

$$\boxed{F = 1.4 \times 10^{3}\ Wm^{-2}}$$

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Q.

Radiant flux Intensity => $1.4 \times 10^{-16}\ Wm^{-2}$ at surface of Earth due to star X

Luminosity of star X => $1.0 \times 10^{30}\ W$

Distance b/w Earth and star X => ?

Ans

$$F = \frac{L}{4\pi d^2}$$

$$1.4 \times 10^{-16} = \frac{1.0 \times 10^{30}}{4 \times 3.14 \times d^2}$$

$$\boxed{d = 2.5 \text{ million ly}}$$

Stellar Radius

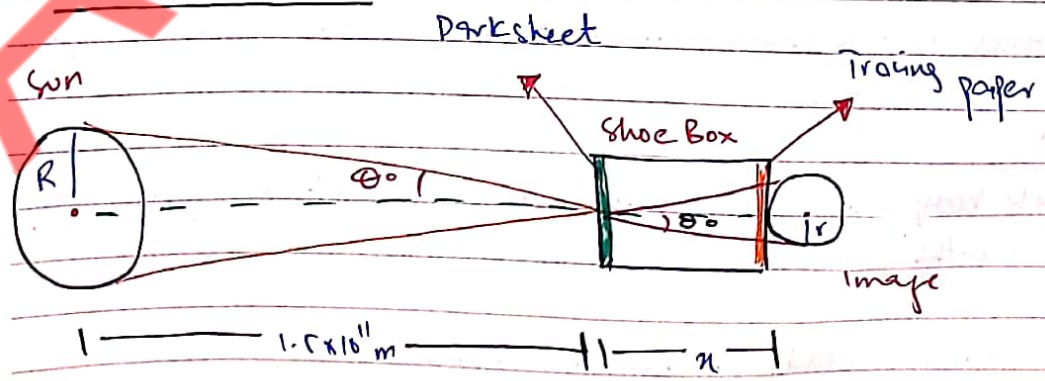
Radius of sun is called "stellar Radius"

→ Generally the radius of star

Experiment

Determination of Radius of Sun

Pin Hole Camera



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→ Place dark [illegible]
→ At this side, using a sharp pin make a tiny hole
→ On the opposite side, place a tracing paper
→ Point the shoe box towards the sun
→ Rays of light enter the shoe box and on the tracing paper image of sun is produced
→ R = Radius of sun
r = radius of image
→ We know the distance of sun from earth = 1.5×10^{11} m
→ We assume x as the distance of shoe box
→ By using tan θ for the ~~right~~ triangle at the left

$$\tan\theta = \frac{R}{1.5\times10^{11}} \quad \text{— i}$$

→ Then for right side's triangle,

$$\tan\theta = \frac{R}{x} \quad \text{— ii}$$

$$\frac{R}{1.5\times10^{11}} = \frac{r}{x}$$ → we can rearrange this

$$R = \frac{r}{x}\times1.5\times10^{11}$$

Black Body idealized

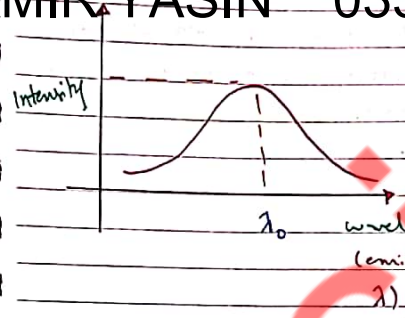
An ~~idealized~~ object which absorbs all the electromagnetic radiations falling on it.

→ It means it absorbs electromagnetic radiations in an infinite range of wavelength

→ Black body emits electromagnetic radiations in infinite range of wavelength

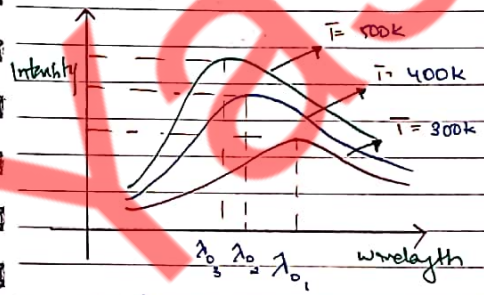
e.g. Sun can be considered as a black body

→ For Sun

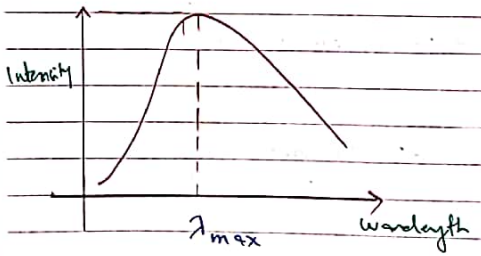


→ Energy emitted by sun, the intensity for all λ are different
→ λ with max intensity is called "λ_0"

For Black Body



→ For certain black body, when we increase the T of the body the λ_0 for max intensity decreases.



$$\lambda_{max} \propto \frac{1}{T}$$

$$\lambda_{max} = \frac{\text{constant}}{T}$$

$$\boxed{\lambda_{max} T = \text{constant}}$$

⇓

mathematical expression for "Wein's Displacement Law"

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Wein's Displacement Law

wavelength of emitted electromagnetic variation for peak intensity is inversely proportional to the temperature of black body.

$$\lambda_{max} T = \text{Constant}$$

Constant = 2.9×10^{-3} mK

Stefan Boltzman Law

→ Luminosity depends on Temp & Area of star

Luminosity $\propto$ (temp)^4

Luminosity $\propto$ Area

$L \propto T^4$ — i

$L \propto A$ — ii

$L \propto AT^4$

$L = (\text{Constant})\, AT^4$

$L = \sigma A T^4$ → mathamatical expression of stefan boltzman law

$A = 4\pi r^2$

r = radius of star

$L = 4\pi\sigma r^2 \times T^4$

σ = boltzman constant (5.67×10^{-8} $W m^{-2} K^{-4}$)

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Q

wavelength of spectrum (lines ~~this~~ observed in Laboratory) = 656.4 nm

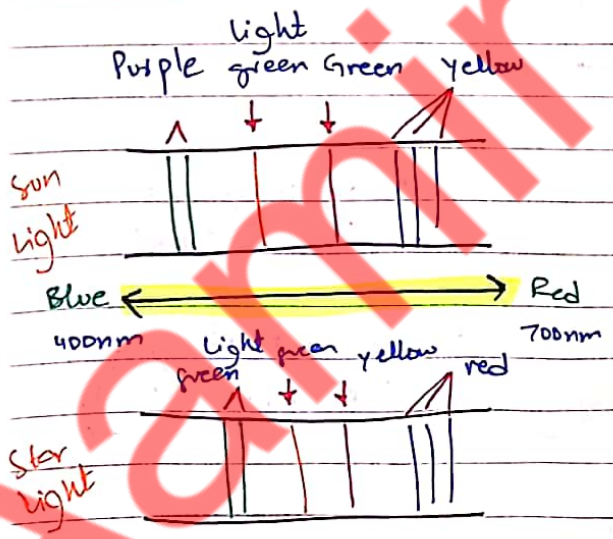
wavelength of same spectrum (lines observed from a distant galaxy) = 663.1 nm

Speed of galaxy => ?

Ans

$$\frac{\Delta x}{\lambda_0} = \frac{v}{c}$$

$c = 3 \times 10^8$ m/s



→ Spectrum produced by sunlight

→ Spectrum produced by any star in the galaxy

* The pattern shifts to the red end, this process is called "red shift"

↳ Indirectly the λ increases, thus frequency decreases ∴ apparent frequency is less than actual frequency thus the star & sun or the both galaxies are moving away from each other

∴ Universe is expanding

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Red Shift / Doppler Shift

A change in the position of the spectrum w.r.to motion of a star / galaxy is called "Red Shift or Doppler's Shift"

* There can never be a blue shift
* The speed with which galaxies are moving away from each other is called "Speed of recession" and it is directly proportional to the distance b/w the galaxies

Hubble's Law

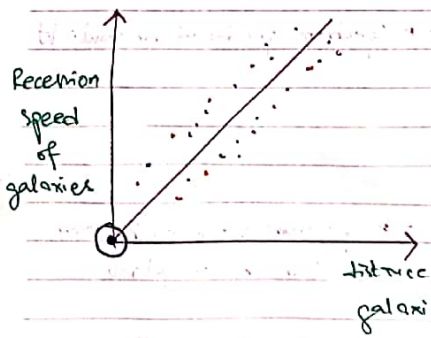
Recession speed of galaxy is directly proportional to the distance of galaxy from us.

Recession speed ∝ Distance

$V \propto d$ $\quad H_0$ = Hubble's Constant

$V = (\text{constant})\, d$ $\quad \Rightarrow 2.4 \times 10^{-18}\ s^{-1}$

$$\boxed{V = H_0 d}$$



> Different galaxies

* Gradient of graph will be equal to H_0

* If we move backward towards the origin, a point will come where the speed & distance was zero that means initially all galaxies were all together => this is evidence of Big Bang Theory.

* It is evident there was a time when all galaxies were at the same position and have moved away from each other

→ The Big Bang theory tells us the creation of universe & space along with time etc

Big Bang Theory

A model of creation of universe from extremely hot & dense state, and its subsequent ~~expansion~~ evolution is called "Big Bang Theory"

$\lambda_{max} T = \text{Constant}$

$T = \dfrac{\text{Constant}}{\lambda_{max}}$ $\quad \Rightarrow$ According to red shift if galaxies are close to each other then λ is less therefore more temp as a evidence to the hot & dense conditions.

→ To find out the time at which big bang happened

speed = $\dfrac{\text{distance}}{\text{time}}$ (speed of recession) $\longrightarrow \boxed{V = H_0 d}$

time = $\dfrac{\text{distance}}{\text{speed}}$

$= \dfrac{9.5 \times 10^{25}\ m}{2.1 \times 10^{7}\ m/s}$

$= 4.52 \times 10^{18}\ s$ => 14 billion years ago

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